



BEST PRACTICE SERIES

D10N/R/T Roller Retainer ARM Coating

**Maintenance
and Repair**

Application

**Component
Life
Management**

**Component
Renewal
(CRC)**

**MARC
Management**

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1.0 Introduction

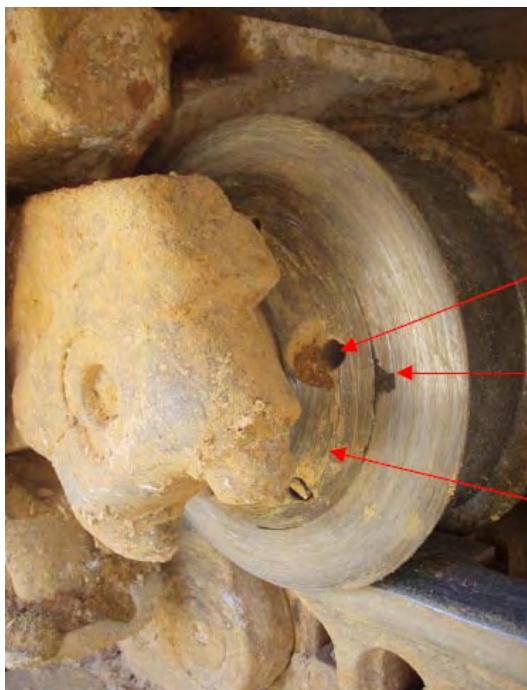
Under normal operating conditions, premature wear can occur on the retainers and bolts for the rollers on the D10T/R/N Track-Type Tractor(s). This wear is due to mixed material flow (abrasive and high impact) across the external face of the rollers. The result of this wear can cause a decrease in roller service life.

Oil leaking from the worn roller retainers and bolts may cause unplanned downtime in the field. In some cases, rollers have to be changed at 50% to 60% of their service life.

This Best Practice will outline a procedure to protect the retainer and bolts from abrasive wear. The objective is to allow 120% of service life on the roller shell without failure in the field. Ultimately, improving roller performance and reducing cost per hour.

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Missing retainer bolt due to abrasive wear.

Oil leaking from the roller due to the loose retainer.

Severe retainer wear to due abrasive material flow.

Figure 1 - Example of wear on the retainer and bolts.

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2.0 Best Practice Description

Some applications for D10N/R/T Track-Type Tractors consist of highly abrasive material. These applications can greatly reduce service life on the rollers, retainers and bolts.

To help increase service life in these applications an Abrasion Resistant Material (ARM) coating may be applied to the roller retainers (both single and double flange). This coating is applied by welding Tungsten Carbide ARM particles on the retainers.

The coating may be applied before installing new rollers or during routine maintenance (shell and seal replacement). It is good practice to coat all rollers at the same time to allow for even wear distribution.

The Tungsten Carbide coating adds 200 grams per roller, which equals a total of 3.2 kg for all 16 rollers. This added weight does not have an affect on equipment operation.

The following figure illustrates Tungsten Carbide ARM coating applied to a roller retainer.

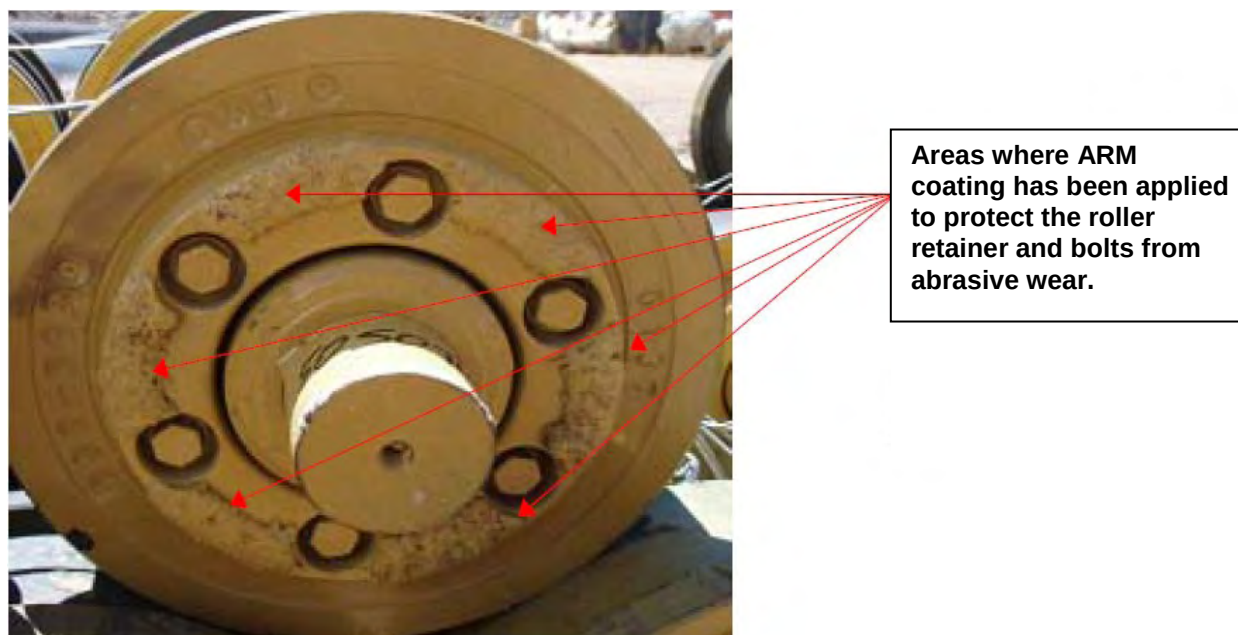


Figure 2 - Tungsten Carbide ARM Coating Applied to the Roller Retainer

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The following picture depicts the roller with Tungsten Carbide ARM coating installed on the undercarriage.

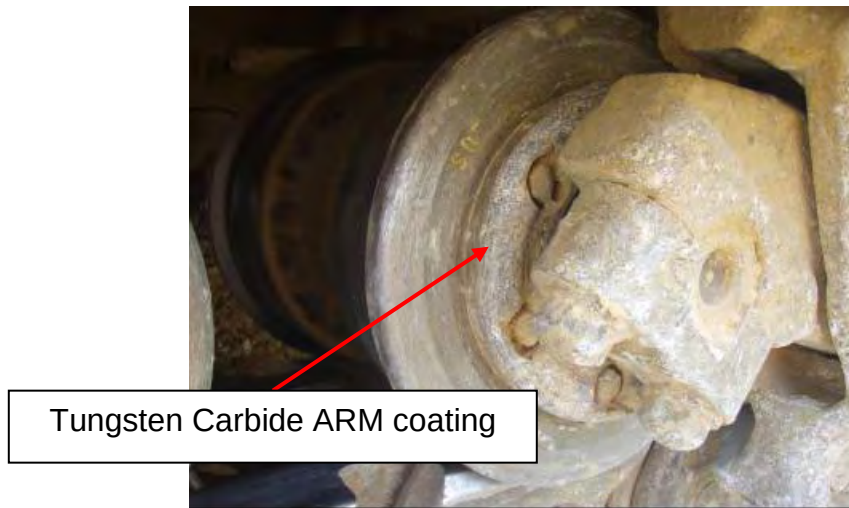


Figure 3 - Roller with Tungsten Carbide ARM coating

3.0 Implementation Steps

In order to properly coat the rollers with Tungsten Carbide ARM, these steps must be followed:

1. Clean the surface of the roller to be coated with Tungsten Carbide ARM using a grinder with steel circular brush attachment. Ensure all paint, rust, debris, etc. is completely removed.
2. Pre-heat the roller to 302 °F (150 °C) using thermal furnace equipment (maintain this temperature while coating with Tungsten Carbide ARM). This allows proper adherence to the roller surface.
3. Weld and/or coat with Tungsten Carbide ARM using a tungsten material dispenser. Ensure to maintain a roller temperature of 302 °F (150 °C).
4. Check your work area to ensure that there are no slag penetration problems.

4.0 Benefits

Benefits associated with this Best Practice include:

- Reduce unscheduled downtime in the field due to leaks or over-heated rollers (dry gaskets).
- Reduce roller cost per hour.
- Increase the service life of the rollers.

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This Best Practice was implemented in a large South American Gold mine. The following three figures demonstrate the benefits realized at this site.

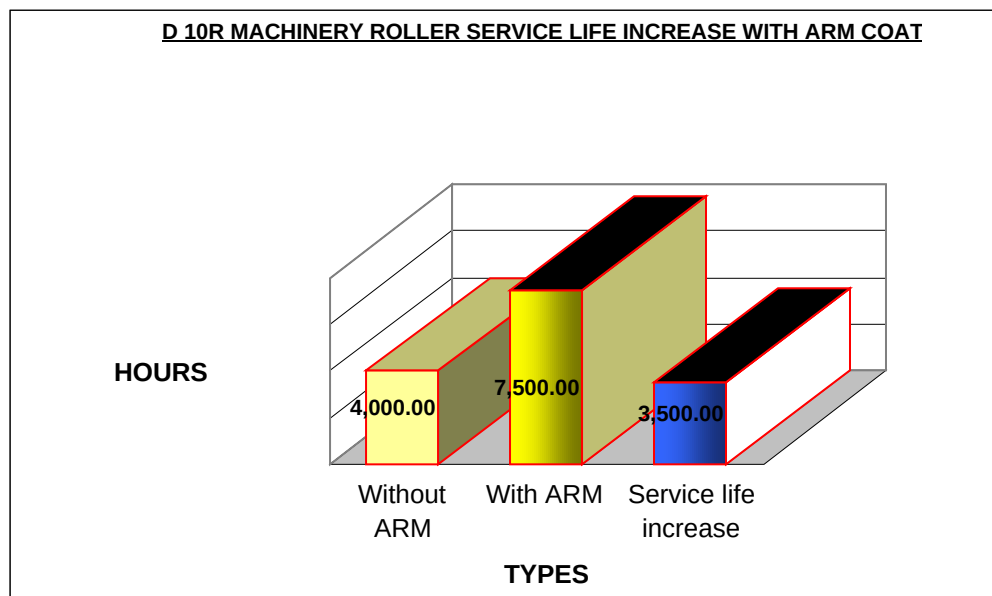


Figure 4 – Example of Roller Service Life Impacts

In this example the service life on the rollers is increased from 4,000 hours to 7,500 hours.

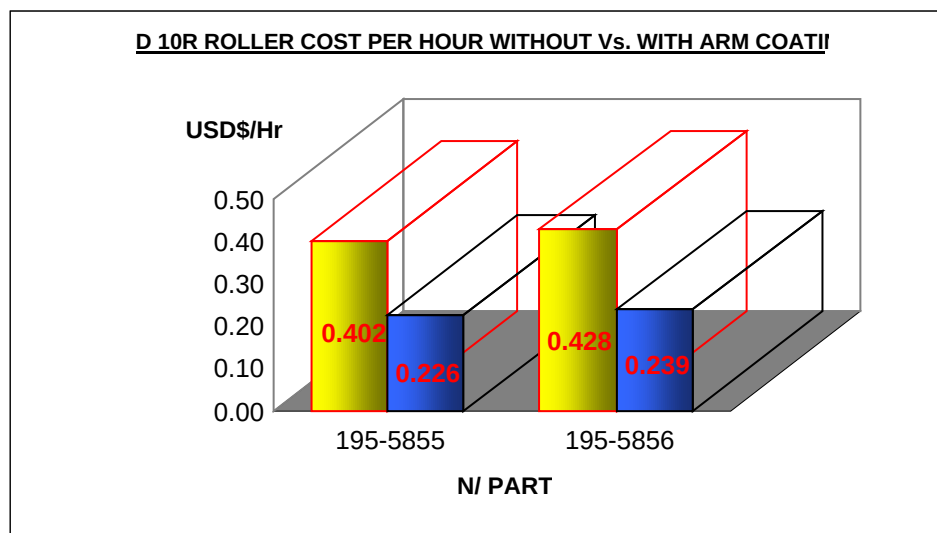


Figure 5 – Example of Roller Cost per Hour Impacts

In the example above, the roller cost per hour decreased by 43.8% for a single flange roller and by 44.2% for a double flange roller.

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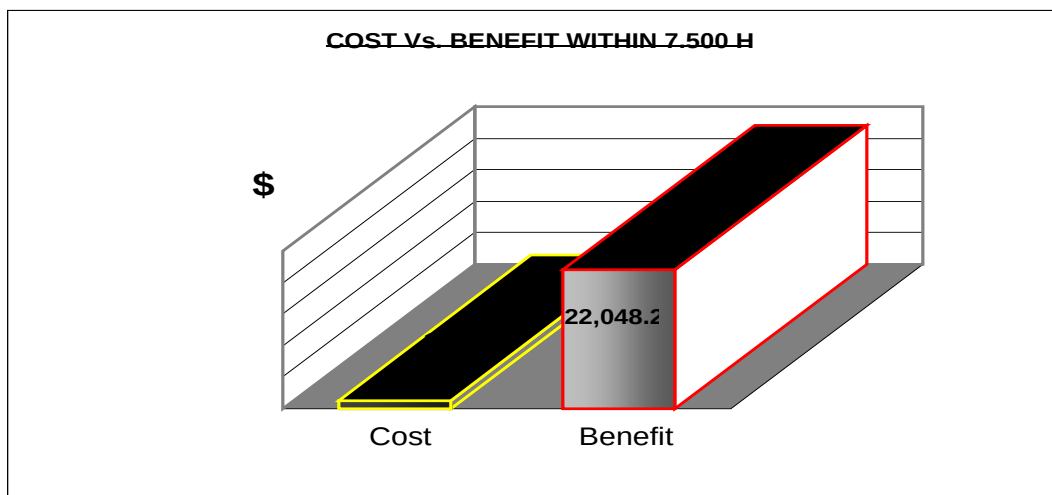


Figure 1 - Example of Cost vs. Benefit Associated with Tungsten Carbide ARM Coating

In the figure above, the benefit realized from coating rollers with Tungsten Carbide ARM is significant compared to the cost. With an investment of \$1,336.00 USD, the total benefit from increasing the service life on 16 rollers to 7,500 hours is \$22,048.20.

5.0 Resources Required

- 3G welder operator, equipment and welding materials.
- Welder labor hours: 2 hours per retainer.
- Undercarriage warehouse coordination.

6.0 Supporting Attachments / References

PEKP9400-04 Custom Track Service Handbook

7.0 Related Best Practices

None

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8.0 Acknowledgements

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